

Multilingual Ticket Translator for Customer Support

Objective

- To automatically detect the language of support tickets, translate them into English for support engineers, and translate the engineer's response back into the customer's original language.

Problem Statement

- Customer support teams receive tickets in multiple regional languages. Agents often rely on manual translation, which is time-consuming and can lead to misunderstandings. This project automates the translation process to improve efficiency and communication.

Technologies Used

- Python
- Language Detection Library (e.g., langdetect)
- Translation API (e.g., Google Translate API)
- File Handling (CSV/TXT/JSON)
- Database (optional)

System Architecture

- Read tickets from a folder.
- Detect the ticket language.
- Translate the ticket to English.
- Store both original and translated versions.
- Allow engineers to reply in English.
- Translate the reply back to the original language.
- Store both reply versions.

Modules

➤ Language Detection Module

- Identifies the source language of each ticket.

➤ Translation Module

- Converts text between the source language and English.

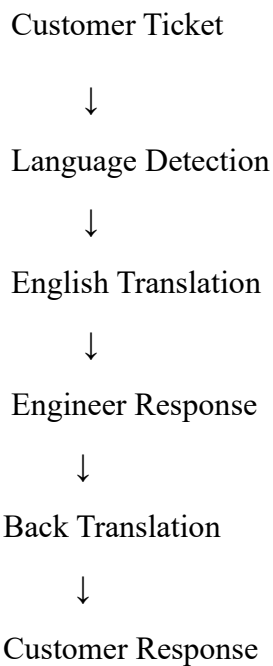
➤ Storage Module

- Stores original and translated tickets and responses.

➤ Response Translation Module

- Translates engineer responses back to the customer's language.

Workflow



Advantages

- Reduces manual translation effort.
- Faster ticket resolution.
- Supports multiple languages.
- Improves communication accuracy.

Future Enhancements

- Custom glossary support.
- Voice ticket translation.
- Real-time chat translation.
- Integration with helpdesk platforms.

Conclusion

- The Multilingual Ticket Translator improves customer support by automating language translation, reducing response time, and enabling seamless communication between customers and support engineers.